Chapter 24: Problems with Solutions

أ.د. زياد عليمات

1. A flat surface of area 3.20 m^2 is rotated in a uniform electric field of magnitude $E = 6.20 \times 10^5$ N/C. Determine the electric flux through this area (a) when the electric field is perpendicular to the surface and (b) when the electric field is parallel to the surface.

For a uniform electric field passing through a plane surface, $\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$, where θ is the angle between the electric field and the normal to the surface.

(a) The electric field is perpendicular to the surface, so $\theta = 0°$:

$$\Phi_E = (6.20 \times 10^5 \text{ N/C})(3.20 \text{ m}^2)\cos 0°$$

$$\Phi_E = \boxed{1.98 \times 10^6 \text{ N} \cdot \text{m}^2/\text{C}}$$

(b) The electric field is parallel to the surface: $\theta = 90°$, so $\cos\theta = 0$, and the flux is $\boxed{\text{zero.}}$

3. **M** A 40.0-cm-diameter circular loop is rotated in a uniform electric field until the position of maximum electric flux is found. The flux in this position is measured to be $5.20 \times 10^5\ \text{N} \cdot \text{m}^2/\text{C}$. What is the magnitude of the electric field?

For a uniform field the flux is $\Phi = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = EA\cos\theta$.

The maximum value of the flux occurs when $\theta = 0$, or when the field is in the same direction as the area vector, which is defined as having the direction of the perpendicular to the area. Therefore, we can calculate the field strength at this point as

$$E = \frac{\Phi_{\text{max}}}{A} = \frac{\Phi_{\text{max}}}{\pi r^2}$$

$$E = \frac{5.20 \times 10^5\ \text{N} \cdot \text{m}^2/\text{C}}{\pi(0.200\ \text{m})^2} = 4.14 \times 10^6\ \text{N/C} = \boxed{4.14\ \text{MN/C}}$$

6. A nonuniform electric field is given by the expression

$$\vec{\mathbf{E}} = ay\hat{\mathbf{i}} + bz\hat{\mathbf{j}} + cx\hat{\mathbf{k}}$$

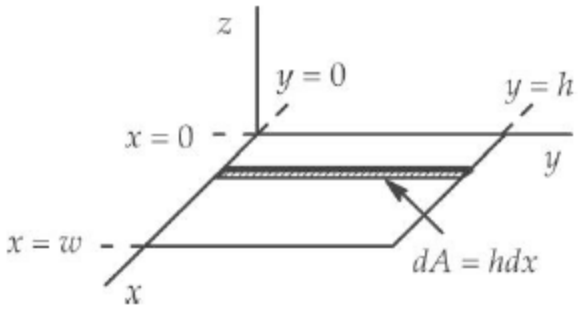
where a, b, and c are constants. Determine the electric flux through a rectangular surface in the xy plane, extending from $x = 0$ to $x = w$ and from $y = 0$ to $y = h$.

We are given an electric field in the general form

$$\vec{\mathbf{E}} = ay\hat{\mathbf{i}} + bz\hat{\mathbf{j}} + cx\hat{\mathbf{k}}$$

In the xy plane, $z = 0$ so that the electric field reduces to

$$\vec{\mathbf{E}} = ay\hat{\mathbf{i}} + cx\hat{\mathbf{k}}$$



ANS. FIG. P24.6

To obtain the flux, we integrate (see ANS. FIG. P24.6 for the definition of dA):

$$\Phi_E = \int \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \int \left(ay\hat{\mathbf{i}} + cx\hat{\mathbf{k}}\right) \cdot \hat{\mathbf{k}}\, dA$$

$$\Phi_E = ch \int_{x=0}^{w} x\, dx = ch \frac{x^2}{2}\bigg|_{x=0}^{w} = \boxed{\frac{chw^2}{2}}$$

Where the $\hat{\mathbf{k}}$ term was eliminated since $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 0$.

7. An uncharged, nonconducting, hollow sphere of radius 10.0 cm surrounds a 10.0-μC charge located at the origin of a Cartesian coordinate system. A drill with a radius of 1.00 mm is aligned along the z axis, and a hole is drilled in the sphere. Calculate the electric flux through the hole.

The electric flux through the hole is given by Gauss's Law (Equation 24.6) as

$$\Phi_{E,\,\text{hole}} = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}}_{\text{hole}} = \left(\frac{k_e Q}{R^2}\right)\left(\pi r^2\right)$$

$$= \left(\frac{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(10.0\times10^{-6}\ \text{C}\right)}{\left(0.100\ \text{m}\right)^2}\right) \times \pi\left(1.00\times10^{-3}\ \text{m}\right)^2$$

$$= \boxed{28.2\ \text{N}\cdot\text{m}^2/\text{C}}$$

8. Find the net electric flux through the spherical closed surface shown in Figure P24.8. The two charges on the right are inside the spherical surface.

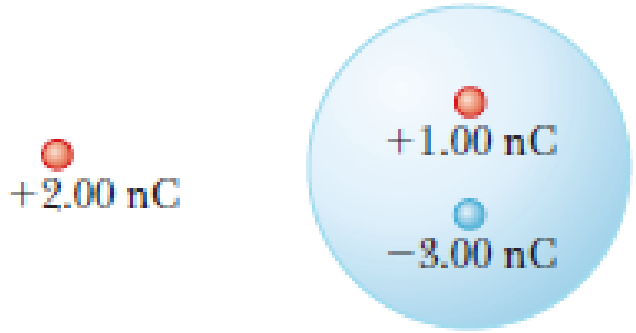


Figure P24.8

The gaussian surface encloses the +1.00-nC and –3.00-nC charges, but not the +2.00-nC charge. The electric flux is therefore

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\left(1.00\times10^{-9}\ \text{C} - 3.00\times10^{-9}\ \text{C}\right)}{8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2} = \boxed{-226\ \text{N}\cdot\text{m}^2/\text{C}}$$

10. The electric field everywhere on the surface of a **W** thin, spherical shell of radius 0.750 m is of magnitude 890 N/C and points radially toward the center of the sphere. (a) What is the net charge within the sphere's surface? (b) What is the distribution of the charge inside the spherical shell?

(a) From $E = \dfrac{k_e Q}{r^2}$,

$$Q = \frac{Er^2}{k_e} = \frac{(8.90 \times 10^2 \text{ N/C})(0.750 \text{ m})^2}{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)} = 5.57 \times 10^{-8} \text{ C}$$

But Q is negative since $\vec{\mathbf{E}}$ points inward, so

$$Q = -5.57 \times 10^{-8} \text{ C} = \boxed{-55.7 \text{ nC}}$$

(b) The [negative] charge has a [spherically symmetric] charge distribution, concentric with the spherical shell.

13. In the air over a particular region at an altitude of 500 m above the ground, the electric field is 120 N/C directed downward. At 600 m above the ground, the electric field is 100 N/C downward. What is the average volume charge density in the layer of air between these two elevations? Is it positive or negative?

Consider as a gaussian surface a box with horizontal area A, lying between 500 and 600 m elevation. From Gauss's Law,

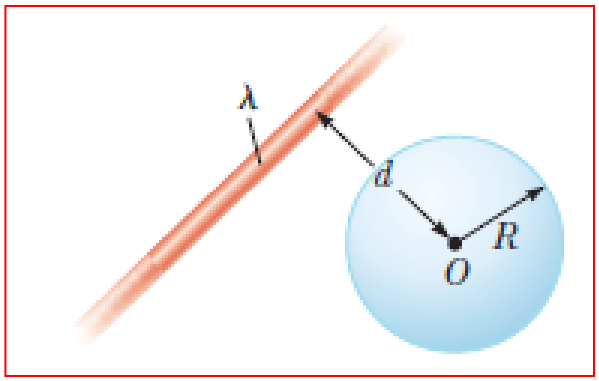
$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q}{\epsilon_0}:$$

$$(+120\ \text{N/C})A + (-100\ \text{N/C})A = \frac{\rho A(100\ \text{m})}{\epsilon_0}$$

$$\rho = \frac{(20.0\ \text{N/C})\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)}{100\ \text{m}} = \boxed{1.77\times10^{-12}\ \text{C/m}^3}$$

The charge is **positive**, to produce the net outward flux of electric field.

17. An infinitely long line charge having a uniform charge per unit length λ lies a distance d from point O as shown in Figure P24.17. Determine the total electric flux through the surface of a sphere of radius R centered at O resulting from this line charge. Consider both cases, where (a) $R < d$ and (b) $R > d$.



(a) If $R \leq d$, the sphere encloses no charge and $\Phi_E = \frac{q_{in}}{\epsilon_0} = \boxed{0}$.

(b) If $R > d$, the length of line falling within the sphere is $2\sqrt{R^2 - d^2}$

so $\Phi_E = \boxed{\frac{2\lambda\sqrt{R^2 - d^2}}{\epsilon_0}}$

19. A particle with charge $Q = 5.00\ \mu\text{C}$ is located at the center of a cube of edge $L = 0.100$ m. In addition, six other identical charged particles having $q = -1.00\ \mu\text{C}$ are positioned symmetrically around Q as shown in Figure P24.19. Determine the electric flux through one face of the cube.

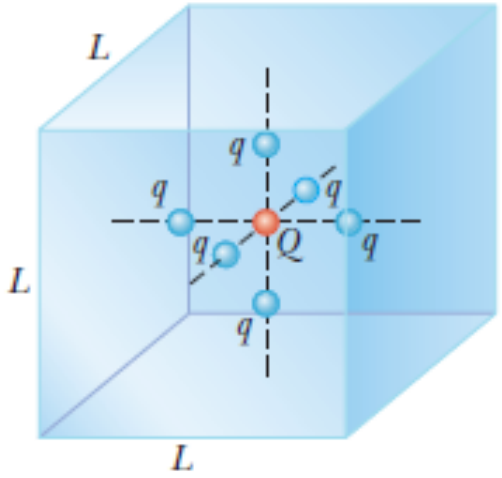


Figure P24.19
Problems 19 and 20.

The total charge is $Q - 6|q|$. The total outward flux from the cube is $\dfrac{Q - 6|q|}{\epsilon_0}$, of which one-sixth goes through each face:

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6\,\epsilon_0}$$

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6\,\epsilon_0} = \frac{(5.00 - 6.00) \times 10^{-6}\ \text{C} \cdot \text{N} \cdot \text{m}^2}{6 \times 8.85 \times 10^{-12}\ \text{C}^2}$$

$$= \boxed{-18.8\ \text{kN} \cdot \text{m}^2 / \text{C}}$$

24. The charge per unit length on a long, straight filament **W** is $-90.0\ \mu\text{C/m}$. Find the electric field (a) 10.0 cm, (b) 20.0 cm, and (c) 100 cm from the filament, where distances are measured perpendicular to the length of the filament.

Note that the electric field in each case is directed radially inward, toward the filament. We use $E = \dfrac{2k_e\lambda}{r}$ and substitute numerical values.

(a) At r = 10.0 cm = 0.100 m,

$$E = \frac{2k_e\lambda}{r} = \frac{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(90.0\times10^{-6}\ \text{C/m}\right)}{0.100\ \text{m}}$$

$$= \boxed{16.2\ \text{MN/C}}$$

(b) At r = 20.0 cm = 0.200 m,

$$E = \frac{2k_e\lambda}{r} = \frac{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(90.0\times10^{-6}\ \text{C/m}\right)}{0.200\ \text{m}}$$

$$= \boxed{8.09\ \text{MN/C}}$$

(c) At r = 100 cm = 1.00 m,

$$E = \frac{2k_e\lambda}{r} = \frac{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(90.0\times10^{-6}\ \text{C/m}\right)}{1.00\ \text{m}}$$

$$= \boxed{1.62\ \text{MN/C}}$$

25. A 10.0-g piece of Styrofoam carries a net charge of AMT $-0.700\ \mu C$ and is suspended in equilibrium above the center of a large, horizontal sheet of plastic that has a uniform charge density on its surface. What is the charge per unit area on the plastic sheet?

The charge per unit area of the plastic sheet must be sufficiently large to result in an upward electric force on the Styrofoam that cancels the downward gravitational force:

$$mg = qE = q\left(\frac{\sigma}{2\epsilon_0}\right) = q\left(\frac{Q/A}{2\epsilon_0}\right)$$

Solving for the charge per unit area gives

$$\frac{Q}{A} = \frac{2\epsilon_0 mg}{q}$$

$$= \frac{2\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left(10.0\times10^{-3}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{\left|-0.700\times10^{-6}\ \text{C}\right|}$$

$$= \boxed{2.48\ \mu\text{C/m}^2}$$

27. M A large, flat, horizontal sheet of charge has a charge per unit area of 9.00 μC/m^2. Find the electric field just above the middle of the sheet.

For a large uniformly charged sheet, $\vec{\mathbf{E}}$ will be perpendicular to the sheet, and will have a magnitude of

$$E = \frac{\sigma}{2\epsilon_0} = 2\pi k_e \sigma$$

$$= (2\pi)\left(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}\right)\left(9.00\times10^{-6}\ \mathrm{C/m^2}\right)$$

so $$\vec{\mathbf{E}} = 5.08\times10^{5}\ \mathrm{N/C}\ \hat{\mathbf{j}}$$

29. M Consider a thin, spherical shell of radius 14.0 cm with a total charge of 32.0 μC distributed uniformly on its surface. Find the electric field (a) 10.0 cm and (b) 20.0 cm from the center of the charge distribution.

(a) Consider the spherical symmetry of the situation. A gaussian sphere concentric wth the shell, with radius 10.0 cm, encloses 0 charge. Then at the surface of this sphere, inside the charged shell, we have $\vec{\mathbf{E}} = \boxed{0}$.

(b) For a gaussian sphere of radius 20.0 cm, we apply $\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{in}}{\epsilon_0}$.

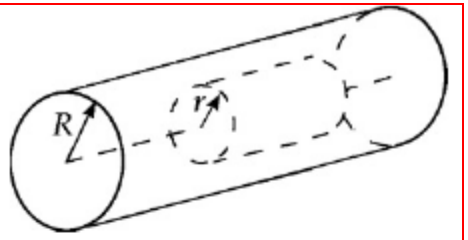
The field is radially outward, and $4\pi r^2 E = q/\epsilon_0$:

$$E = \frac{k_e q}{r^2} = \frac{\left(8.99\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(32.0\times10^{-6}\ \text{C}\right)}{(0.200\ \text{m})^2}$$

$$= 7.19\times10^6\ \text{N/C}$$

33. Consider a long, cylindrical charge distribution of radius R with a uniform charge density ρ. Find the electric field at distance r from the axis, where $r < R$.

If ρ is positive, the field must be radially outward. Choose as the gaussian surface a cylinder of length L and radius r, contained inside the charged rod. Its volume is $\pi r^2 L$ and it encloses charge $\rho\pi r^2 L$. Because the charge distribution is long, no electric flux passes through the circular end caps; $\vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = EdA\cos 90.0° = 0$. The curved surface has $\vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = EdA\cos 0°$, and E must be the same strength everywhere over the curved surface.



ANS. FIG. P24.33

Gauss's law, $\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \dfrac{q}{\epsilon_0}$, becomes $E \int_{\text{Curved Surface}} dA = \dfrac{\rho\pi r^2 L}{\epsilon_0}$.

Now the lateral surface area of the cylinder is $2\pi rL$:

$$E(2\pi r)L = \frac{\rho\pi r^2 L}{\epsilon_0}$$

Thus, $\vec{\mathbf{E}} = \boxed{\dfrac{\rho r}{2\epsilon_0} \text{ radially away from the cylinder axis}}$.

37. M A long, straight metal rod has a radius of 5.00 cm and a charge per unit length of 30.0 nC/m. Find the electric field (a) 3.00 cm, (b) 10.0 cm, and (c) 100 cm from the axis of the rod, where distances are measured perpendicular to the rod's axis.

$\oint E\,dA = E(2\pi rl) = \frac{q_{in}}{\epsilon_0}$ $\quad E = \frac{q_{in}/l}{2\pi\epsilon_0 r} = \frac{\lambda}{2\pi\epsilon_0 r}$ for the field outside the metal rod.

(a) At $r = 3.00$ cm, $\quad \vec{\mathbf{E}} = \boxed{0}$

(b) At $r = 10.0$ cm,

$$\vec{\mathbf{E}} = \frac{30.0\times10^{-9}\ \text{C}}{2\pi\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)(0.100\ \text{m})}$$

$$= \boxed{5\,400\ \text{N/C, outward}}$$

(c) At $r = 100$ cm,

$$\vec{\mathbf{E}} = \frac{30.0\times10^{-9}\ \text{C}}{2\pi\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)(1.00\ \text{m})}$$

$$= \boxed{540\ \text{N/C, outward}}$$

54. GP A solid, insulating sphere of radius a has a uniform charge density throughout its volume and a total charge Q. Concentric with this sphere is an uncharged, conducting, hollow sphere whose inner and outer radii are b and c as shown in Figure P24.54 (page 744). We wish to understand completely the charges and electric fields at all locations. (a) Find the charge contained within a sphere of radius $r < a$. (b) From this value, find the magnitude of the electric field for $r < a$. (c) What charge is contained within a sphere of radius r when $a < r < b$? (d) From this value, find the magnitude of the electric field for r when $a < r < b$. (e) Now consider r when $b < r < c$. What is the magnitude of the electric field for this range of values of r? (f) From this value, what must be the charge on the inner surface of the hollow sphere? (g) From part (f), what must be the charge on the outer surface of the hollow sphere? (h) Consider the three spherical surfaces of radii a, b, and c. Which of these surfaces has the largest magnitude of surface charge density?

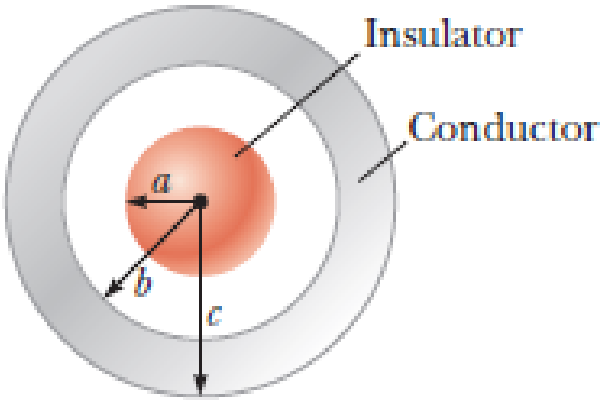
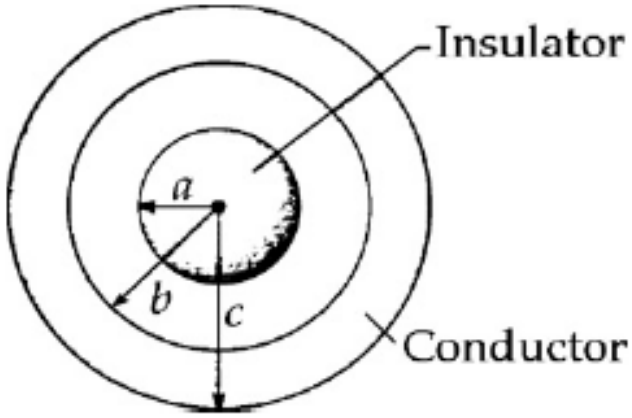


Figure P24.54
Problems 54, 55, and 57.

Choose as each gaussian surface a concentric sphere of radius r. The electric field will be perpendicular to its surface, and will be uniform in strength over its surface. The density of charge in the insulating sphere is

$$\rho = Q / \left(\tfrac{4}{3}\pi a^3\right)$$



ANS. FIG. P24.54

(a) The sphere of radius $r < a$ encloses charge

$$q_{\text{in}} = \rho\left(\frac{4}{3}\pi r^3\right) = \left(\frac{Q}{\frac{4}{3}\pi R^3}\right)\left(\frac{4}{3}\pi r^3\right) = \boxed{Q\left(\frac{r}{R}\right)^3}$$

(b) Applying Gauss's law to this sphere reveals the magnitude of the field at its surface.

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0}\left(\frac{r}{a}\right)^3 \rightarrow E = \frac{1}{4\pi\,\epsilon_0}\frac{Qr}{a^3} = \boxed{k_e \frac{Qr}{a^3}}$$

(c) For a sphere of radius r with $a < r < b$, the whole insulating sphere is enclosed, so the charge within is Q: $q_{\text{in}} = \boxed{Q}$.

(d) Gauss's law for this sphere becomes:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0} \rightarrow E = \frac{1}{4\pi\,\epsilon_0}\frac{Q}{r^2} = \boxed{k_e \frac{Q}{r^2}}$$

(e) For $b \le r \le c$, $\boxed{E = 0}$ because there is no electric field inside a conductor.

(f) For $b \le r \le c$, we know $E = 0$. Assume the inner surface of the hollow sphere holds charge Q_{inner}. By Gauss's law,

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$0 = \frac{Q + Q_{\text{inner}}}{\epsilon_0} \rightarrow Q_{\text{inner}} = \boxed{-Q}$$

(g) The total charge on the hollow sphere is zero; therefore, charge on the outer surface is opposite to that on the inner surface:

$$Q_{\text{outer}} = -Q_{\text{inner}} = \boxed{+Q}$$

(h) A surface of area A holding charge Q has surface charge $\sigma = q/A$. The solid, insulating sphere has small surface charge because its total charge Q is uniformly distributed throughout its volume. The inner surface of radius b has smaller surface area, and therefore larger surface charge, than the outer surface of radius c.